

Mathematics for Computing

Set Theory

Gokul M

Department of Computer Science

Semester I

M.Tech in Data Science and Artificial Intelligence

Cochin University of Science and Technology (CUSAT)

July 2026

Mathematical Forms of Sets

$$\textcircled{1} \quad S = \{3, 6, 9, 12, \dots\}$$
$$S = \{x \in \mathbb{N} \mid 3 \mid x\}$$

$$\textcircled{2} \quad S = \{1, 3, 5, 7, \dots\}$$
$$S = \{x \in \mathbb{N} \mid x = 2n - 1\}$$

$$\textcircled{3} \quad S = \{2, 4, 6, 8, \dots\}$$
$$S = \{x \in \mathbb{N} \mid x = 2n\}$$

$$\textcircled{4} \quad S = \{(1, 2), (2, 4), (3, 6), \dots\}$$
$$S = \{(x, y) \mid y = 2x, x, y \in \mathbb{N}\}$$

$$\textcircled{5} \quad S = \{1, 4, 9, 16, \dots\}$$
$$S = \{x \mid x = n^2, n \in \mathbb{N}\}$$

$$\textcircled{6} \quad S = \{1, 8, 27, 64, \dots\}$$
$$S = \{x \mid x = n^3, n \in \mathbb{N}\}$$

7 $S = \{1, 2, 6, 24, \dots\}$
 $S = \{x \mid x = n!, n \in \mathbb{N}\}$

8 $S = \{4, 7, 10, 13, \dots\}$
 $S = \{x \mid x = 3n + 1, n \in \mathbb{N}\}$

9 $S = \{5, 10, 15, 20, \dots\}$
 $S = \{x \mid x = 5n, n \in \mathbb{N}\}$

10 $S = \{2, 4, 8, 16, \dots\}$
 $S = \{x \mid x = 2^n, n \in \mathbb{N}\}$

Types of Sets

- **Cardinality of a Set**

- Number of elements in a set.
- AI Example: Image classes $A = \{\text{Cat}, \text{Dog}, \text{Car}, \text{Tree}\}$, so $|A| = 4$.

- **Finite Set**

- A set having a limited number of elements.
- AI Example: Facial expression labels $\{\text{Happy}, \text{Sad}, \text{Angry}, \text{Neutral}\}$.

- **Infinite Set**

- A set having infinitely many elements.
- AI Example: All possible real-valued temperature sensor readings.

- **Null (Empty) Set**

- A set containing no elements.
- AI Example: Images that are both "Cat" and "Car" simultaneously: $\emptyset$.

- **Equal Sets**

- Two sets containing exactly the same elements.
- AI Example: Model A detects {Cat, Dog} and Model B detects {Dog, Cat}.

- **Equivalent Sets**

- Two sets having the same number of elements.
- AI Example: {Cat, Dog, Bird} and {Car, Bus, Bike}.

- **Subset**

- Every element of one set belongs to another set.
- AI Example: Dog images $\subseteq$ Animal images.

- **Superset**

- A set containing all elements of another set.
- AI Example: Animal dataset $\supseteq$ Dog dataset.

- **Universal Set**

- A set containing all objects under consideration.
- AI Example: Complete ImageNet image collection.

- **Power Set**

- Set of all possible subsets of a set.
- AI Example: {Age, Income} has $\{\emptyset, \{\text{Age}\}, \{\text{Income}\}, \{\text{Age}, \text{Income}\}\}$.

- **Singleton Set**

- A set containing exactly one element.
- AI Example: Predicted class = {Cat}.

- **Countable Set**

- A set whose elements can be counted one by one.
- AI Example: User IDs in an AI recommendation system.

Inclusion–Exclusion Principle

Definition

The Inclusion–Exclusion Principle is used to find the number of elements in the union of two or more overlapping sets.

Formula for Two Sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

The General Formula for n Sets

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left| \bigcap_{i=1}^n A_i \right|$$

Relations

Definition

A relation R on a set A is a subset of the Cartesian product $A \times A$.

$$R \subseteq A \times A$$

If $(a, b) \in R$, we write

$$aRb$$

which means that a is related to b .

Example

Let

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}.$$

Partially Ordered Sets

Definition

A partially ordered set (Poset) is a set together with a relation that satisfies the following properties:

- **Reflexive:**

$$\forall a \in A, aRa$$

- **Antisymmetric:**

$$aRb \text{ and } bRa \implies a = b$$

- **Transitive:**

$$aRb \text{ and } bRc \implies aRc$$

Example 1

Set

$$A = \{1, 2, 3, 4, 5\}$$

Relation

$$aRb \iff a \leq b$$

Hasse Diagram



Example 2

Set

$$A = \{1, 2, 3\}$$

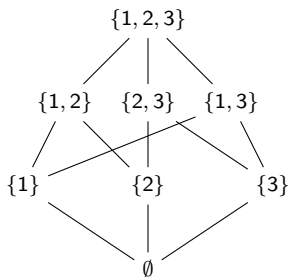
Relation

$$R = (2^A, \subseteq)$$

where

$$2^A = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Hasse Diagram



Example 3

Set

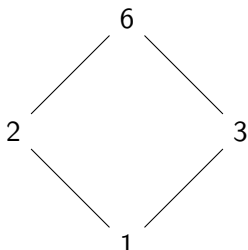
$$A = \{1, 2, 3, 6\}$$

Relation

$$aRb \iff a \mid b$$

where " $a \mid b$ " means " a divides b ".

Hasse Diagram



Chains and Antichains

Chain

A chain is a subset of a partially ordered set in which every pair of elements is comparable.

$$a \leq b \quad \text{or} \quad b \leq a$$

for every pair a, b in the chain.

Example

For

$$A = \{1, 2, 3, 4, 5\}$$

with the usual ordering $\leq$,

$$\{1, 2, 3, 4, 5\}$$

is a chain.

Chains and Antichains

Antichain

An antichain is a subset in which no two distinct elements are comparable.

Example

Consider the partially ordered set

$$A = \{1, 2, 3, 4, 6, 12\}$$

with the divisibility relation

$$aRb \iff a \mid b.$$

Chain: $\{1, 2, 6, 12\}$

Since $1 \mid 2$, $2 \mid 6$, $6 \mid 12$, every pair of elements is comparable.

Antichain: $\{3, 4\}$

Since $3 \nmid 4$ and $4 \nmid 3$, the elements are incomparable.